

AYOLEYI GBENGA-AYODEJI

MARVELOUS

Lagos/Ogun State, Nigeria • github.com/Ayoleyi-dev • ayoleyi-s-digital-architecture-hub.vercel.app

Biochemistry undergraduate and computational researcher specializing in computer-aided drug discovery (CADD) and structural biology. Experienced in bridging wet-lab principles with dry-lab execution, utilizing Linux/WSL environments to run molecular docking simulations, perform homology modeling, and analyze large-scale genomic datasets.

EDUCATION

Bachelor of Science in Biochemistry

Expected Aug 2027

University of Lagos (UNILAG)

RESEARCH & COMPUTATIONAL EXPERIENCE

AI-Driven Homology & Targeted Drug Discovery Project

June 2026

Independent Academic Research

- Executed sequence retrieval, alignment, and 3D homology modeling using SWISS-MODEL and Biopython to predict protein structures.
- Conducted molecular docking simulations utilizing AutoDock Vina to evaluate binding affinities and screen compound libraries.
- Visualized protein-ligand interactions using PyMOL to interpret binding energy, spatial conformations, and structural viability.

Laboratory Intern (SIWES)

2025

National Agency for Food and Drug Administration and Control (NAFDAC)

- Supported quality assurance and regulatory compliance processes within the laboratory, ensuring adherence to standard operating procedures.

PROJECTS

Agrospectra: NDVI Crop Health Analysis

- Developed a Python remote sensing pipeline utilizing NumPy and OpenCV to calculate and visualize crop health indices (NDVI) from spectral arrays.

TECHNICAL SKILLS

- Bioinformatics & CADD:** Homology Modeling, Molecular Docking (AutoDock Vina), Sequence Alignment, Binding Energy Interpretation, Chemoinformatics.
- Scientific Tools:** PyMOL, Modeller / SWISS-MODEL, RDKit, Biopython.
- Programming & Environment:** Python (NumPy, Pandas), R, SQL, Linux / WSL (Ubuntu), Bash Scripting, Git/GitHub.

CERTIFICATIONS

- Professional Certificate: Bridging Bioinformatics and Chemoinformatics in Automated Drug Discovery (Started June 2026)